

Baseline taper:

Initial Concentration : $C_i = 5\%$ (50 mg/mL)

Final Concentration : $C_f = 0.5\%$ (5 mg/mL)

Total Span : $T_{\text{weeks}} = 12$ to 24 weeks

$$\Delta C_{\text{standard}} = \frac{C_i - C_f}{T_{\text{weeks}}}$$

$$\begin{aligned} \text{Ex: } \Delta C_{\text{standard}} &= \frac{5 - 0.5}{12} \\ &= 0.375\% \text{ decrease} \\ &\quad \text{a week} \end{aligned}$$

Average Benchmark and Intensity:

* Numbers come from Research paper sent.
Standard puff duration: 2 seconds

Average: 172 puffs

* Daily Duration (Days)

* Usage Multiplier (M_u)

$D_{avg} = < 240s$, then $M_u = 0.9$. Speeds up taper slightly for high compliance.

$D_{avg} = 240s - 450s$, then $M_u = 1$. Taper stays on track.

$D_{avg} = 450s - 700s$, then $M_u = 1.1$. Slows the taper to prevent craving spikes.

$D_{avg} = > 700s$, then $M_u = 1.3$. Strong break, indicates the user is struggling.

Qualitative Intake:

* Subjective Stress Value (S): $[0.5, -1]$

Survey Answer	Value (S)	
Feeling Great/ No Cravings	+0.5	High success prob, minor acceleration
Manageable/ Minor Cravings	+0	Standard progress
Stressed/Thinking about Nicotine	-0.5	Warning, Slows down
Relapsed/Used a cig	-1	Failure State, plateau or slight reversal

Final adjustment:

Every Sunday, C_{next} is calculated.

* Adaptive Multiplier (A): $A = (1+S) + (1-M_u)$

$$C_{next} = C_{current} - (\Delta C_{std} \cdot A)$$

* Note that if C_{next} exceeds the initial concentration of 5%, it will be capped at 5%.

$$V_{nic} = 30 \cdot (C_{next} / 5.0)$$

$$V_{dil} = 30 - V_{nic}$$

61-68: "Growth Zones"

51-58: "Stability Zones"

A		B		C		D		E		F		G		H	
Scenario_Drop_Rates															
ID		#	Usage (Mu)		Survey (S)		Multiplier (A)		Weekly %(3 Month)		Weekly %(4 Month)		Weekly %(6 Month)		Clinical Description
G1			0.9		0.5		1.6		0.6		0.45		0.3		PATIENT EXCELLENCE: High self-efficacy and low dependency.
G2			0.9		0		1.1		0.41		0.31		0.21		EFFICIENT PROGRESS: Physical usage low; patient is stable.
G3			0.9		-0.5		0.6		0.23		0.17		0.11		MENTAL FATIGUE OVERRIDE: Prioritizes mental stability.
G4			0.9		-1		0.1		0.04		0.03		0.02		PSYCHOLOGICAL PLATEAU: 90% slow-down to prevent relapse.
G5			1		0.5		1.5		0.56		0.42		0.28		CONFIDENCE BOOST: On-track usage with high morale.
G6			1		0		1		0.38		0.28		0.19		THE LINEAR ANCHOR: Perfect alignment; standard baseline.
G7			1		-0.5		0.5		0.19		0.14		0.09		CAUTIOUS DAMPENING: Cuts speed in half for adjustment.
G8			1		-1		0		0		0		0		SAFETY HOLD: Taper paused to provide consistent nicotine.
S1			1.1		0.5		1.4		0.53		0.39		0.27		COMPENSATORY OFFSET: Progress while usage is rising.
S2			1.1		0		0.9		0.34		0.25		0.17		MILD RESISTANCE: Slight 10% safety buffer introduced.
S3			1.1		-0.5		0.4		0.15		0.11		0.08		WITHDRAWAL WARNING: Taper shifts to micro-step.
S4			1.1		-1		-0.1		-0.04		-0.03		-0.02		RECOVERY STEP (MINOR): Slight nicotine increase for relapse.
S5			1.3		0.5		1.2		0.45		0.34		0.23		HEAVY COMPENSATORY OFFSET: Restricted pace for heavy usage.
S6			1.3		0		0.7		0.26		0.2		0.13		HEAVY DEPENDENCE DEFENSE: Forces significant slow-down.
S7			1.3		-0.5		0.2		0.08		0.06		0.04		CRITICAL MICRO-STEP: Minimal progress to maintain habit.
S8			1.3		-1		-0.3		-0.11		-0.08		-0.06		RESCUE STABILIZATION: Emergency brake; stabilization dose.